ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** fat-rich food	Bile aids fat digestion.
2	**(c)** 1,15,200 times	80 x 60 x 24.
3	**(d)** to provide a large surface area for absorption	Large area.
4	**(d)** digestion of all three food types	Pancreatic juice is important.
5	**(a)** 8000 mL	16 x 500.
6	**(a)** plant food takes longer to digest	Cellulose.
7	**(c)** poor protein digestion and more germs reaching the intestine	HCl needed.
8	**(c)** 480 litres	8 L x 60.
9	**(c)** shorter	Meat is easier to digest.
10	**(a)** poor absorption of digested food	Less surface area.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Respiration releases energy as heat (1); Metabolism increases after a meal (1).	1+1
12	Right sends blood to lungs nearby (1); Left pumps to the whole body (1).	1+1
13	Higher concentration in alveoli (1); Diffusion (1).	1+1
14	Muscles need more oxygen (1); Blood must carry it faster (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fat digestion poor (1); Fewer enzymes (1); Special diet needed (1).	1+1+1
16	Move to site (1); Engulf germs (1); Defence (1).	1+1+1
17	Adds bulk (1); Speeds movement (1); Prevents constipation (1).	1+1+1
	OR: Muscles need more oxygen (1); Heart pumps more (1); Blood removes CO2 faster (1).	1+1+1
18	Insulin is low (1); Blood sugar rises (1); Control diet (1).	1+1+1
	OR: Frog uses oxygen (1); Needs surface air (1); Becomes distressed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rest: count breaths per minute (1); After walking (1); After running (1); Table of results (1); Rate rises with activity (1).	1+1+1+1+1
	OR: Lung damage (1); Blood vessel damage (1); Heart disease (1); Toxins strain kidneys (1); Avoid smoking (1).	1+1+1+1+1
20	Right side to lungs (1); Oxygenation (1); Back to left heart (1); Pumped to body (1); Passes through heart twice (1).	1+1+1+1+1
	OR: Human: lungs, air (1); Fish: gills, water (1); Insect: tracheae (1); Efficiency differs (1); Adaptation to habitat (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10 million	5 x 2.
2	**(c)** the iron-containing haemoglobin	Iron.
3	**(c)** inhalation is affected	Breathing needs it.
4	**(d)** 10.5 L	1.5 x 7.
5	**(c)** arteries have thick walls to bear high pressure	Pressure.
6	**(d)** thin air has less oxygen	Compensation.
7	**(d)** 240 hours	8 x 30.
8	**(d)** carbon dioxide	Gas exchange.
9	**(c)** carbon dioxide has built up	CO2 triggers breathing.
10	**(d)** 0.83 s	60/72.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Millions of alveoli (1); Large surface for gas exchange (1).	1+1
12	Adds bulk (1); Prevents constipation (1).	1+1
13	Fats would not digest properly (1); Poor absorption of fatty acids (1).	1+1
14	Dilutes urine (1); Helps pass stones (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sugar is digested quickly (1); Blood sugar drops (1); Lack of fibre and protein (1).	1+1+1
16	Waves of muscle contraction (1); Moves food along (1); Works in any posture (1).	1+1+1
17	Respiration gives heat (1); Sweat evaporates (1); Blood vessels adjust (1).	1+1+1
	OR: Dehydration (1); Concentrated wastes (1); Drink more water (1).	1+1+1
18	Training (1); Stronger muscles (1); More oxygen use (1).	1+1+1
	OR: Contains antibodies (1); Easy to digest (1); Infants lack enzymes for solid food (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rice starch to sugar (1); Dal protein to amino acids (1); Vegetables: fibre for egestion (1); Absorbed by villi (1); Used for energy and growth (1).	1+1+1+1+1
	OR: Filter blood (1); Reabsorb water (1); Remove urea (1); Dialysis machine cleans blood (1); For kidney failure (1).	1+1+1+1+1
20	Digestive supplies glucose (1); Respiratory supplies oxygen (1); Circulatory transports both (1); Cells release energy (1); Wastes removed (1).	1+1+1+1+1
	OR: Press to stop flow (1); Platelets clot (1); Clean wound (1); Cover (1); See a doctor (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** carbon dioxide	Gas exchange in tissues.
2	**(d)** partial breakdown of glucose without oxygen forms lactic acid	Anaerobic respiration.
3	**(d)** beats less often but pumps more per beat	Efficient.
4	**(d)** liver	Made from proteins.
5	**(d)** heart, aorta, arteries, capillaries, veins, heart	Correct circuit.
6	**(a)** aorta, arteries, capillaries	Decreasing size.
7	**(a)** water, urea and salts	Waste.
8	**(a)** remove wastes from blood	Artificial kidney.
9	**(c)** 6 to 7 m	Long coiled tube.
10	**(d)** teeth	Teeth are mechanical.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Damages the lining of the food pipe and stomach (1); Causes acidity (1).	1+1
12	Blood goes to the digestive organs (1); Muscles may get less blood, causing cramps (1).	1+1
13	Cells need constant supply (1); Wastes must be removed (1).	1+1
14	Aerobic uses oxygen (1); Anaerobic occurs without oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	More oxygen needed (1); Carbon dioxide removed (1); Larger volume per breath (1).	1+1+1
16	Oxygen-rich from lungs (1); Oxygen-poor from body (1); Do not mix (1).	1+1+1
17	Less oxygen carrying (1); Muscles need more (1); Breathing speeds up (1).	1+1+1
	OR: Stomach acidic (1); Intestine alkaline (1); Different enzymes act (1).	1+1+1
18	One-cell thick wall (1); Easy exchange (1); Close to every cell (1).	1+1+1
	OR: Germs (1); Cause diarrhoea (1); Contaminated water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Herbivore flat teeth (1); Herbivore long intestine (1); Carnivore sharp teeth (1); Short intestine (1); Different digestion (1).	1+1+1+1+1
	OR: Acidity: spicy food (1); Ulcers: acid damage (1); Constipation: low fibre (1); Diarrhoea: germs (1); Hygiene and diet (1).	1+1+1+1+1
20	Tiredness (1); Infections (1); Iron-rich diet (1); Medical advice (1); Rest (1).	1+1+1+1+1
	OR: Food has chemical energy (1); Digestion gives glucose (1); Respiration releases energy (1); Used for movement (1); Excess stored as fat (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.